

1. Overview

This document covers the complete setup of Jupyter Notebook using Docker. Docker allows us to run Jupyter in isolated container without installing it directly on our machine.

2. Step-by-Step Setup

Step 1 — Install Docker and Python

Step 2 — Create Project Folder

Step 3 — Create Dockerfile

Write this code inside dockerfile –

```
FROM python:3.11-slim

WORKDIR /app

RUN pip install jupyter notebook

EXPOSE 8888

RUN mkdir -p /app/notebooks

CMD ["jupyter", "notebook", "--ip=0.0.0.0", "--allow-root", "--NotebookApp.token=", "--NotebookApp.password="]
```

Step 4 — Build Docker Image

Run this command inside the folder where Dockerfile is saved:

```
docker build -t jupyter-notebook .
```

Step 5 — Run the Container

```
docker run -d -p 8888:8888 -v <desktop-folder-directory>:/app/notebooks --name my-jupyter jupyter-notebook
```

```
docker start my-jupyter
```

Step 6 — Open in Browser

Open your browser and go to:

```
http://localhost:8888
```

Step 7 — Run Sample Notebook

Click 'New' > 'Python 3' to create a new notebook. Run the following code to verify the setup:

```
>> pip install numpy matplotlib

import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y)
plt.title('Sample Graph docker Jupyter')
plt.show()
```

3. Common Issues & Solutions

Issue	Solution
Port already in use	Change -p 8889:8888 and open localhost:8889
Container name already exists	Run: docker rm my-jupyter, then re-run
Packages not found	Add them to Dockerfile RUN pip install line and rebuild